

THE MAXIMUM EULER–SOMBOR INDEX AT DIAMETER THREE, AND BOTH EXTREMA AT NINE VERTICES

ABSTRACT. The Euler–Sombor index of a graph G is

$$EU(G) = \sum_{xy \in E(G)} \sqrt{d(x)^2 + d(x)d(y) + d(y)^2}.$$

Khanra and Das asked for the extremal values and extremal graphs of EU over “the class of all connected graphs with given diameter”; Sekar, Balachandran, Elumalai and Liu reproduce the request as their “Problem 1”, and a 2026 survey by Ali, Gutman, Réti, Albalahi and Hamza poses it with the order fixed as well, which is the reading taken here. We settle one column and one cell, not the problem. For every $n \geq 4$ the maximum of EU over the connected graphs of order n and diameter exactly 3 equals

$$\frac{\sqrt{3}}{2}(n-2)^2(n-3) + \sqrt{n^2 - 3n + 3} + (n-3)\sqrt{3n^2 - 15n + 19},$$

attained by exactly one graph up to isomorphism, namely $H(1, n-3)$; the proof is a closed form and uses no computation. We also settle the single cell $n = 9$, $D = 3$ in the other direction: over the 148 229 isomorphism classes of that cell the minimum is $8\sqrt{3} + 6\sqrt{19} = 40.0098\dots$, attained by exactly one class, the *bicyclic* theta graph $\Theta(3, 3, 4)$, so that the minimiser there is neither a tree nor unicyclic and the best tree in the cell is only third. That half is settled by the exhaustive census shipped with this paper, not by hand. Problem 1 quantifies over all n and all D and is *not* resolved here; §6 says exactly what remains, including the one paywalled paper that could reduce our maximum half to a rediscovery.

1. THE PROBLEM AND ITS SOURCE

For a graph G write $d(x) = d_G(x)$. The *Euler–Sombor index* is

$$(1) \quad EU(G) = \sum_{xy \in E(G)} \sqrt{d(x)^2 + d(y)^2 + d(x)d(y)},$$

in the normalisation printed on journal page 761 of [5] and in the abstract of [3]. Khanra and Das close [2] with a list of open problems whose item (1) reads, on journal page 546, verbatim:

Here we propose the following open problems on Euler Sombor index of a graph: Determine the extremal values and extremal graphs with respect to Euler Sombor index over

2020 *Mathematics Subject Classification.* 05C09, 05C35, 05C12, 92E10.

Key words and phrases. Euler–Sombor index, degree-based topological index, diameter, extremal graph, theta graph, bicyclic graph.

the following sets: (1) the class of all connected graphs with given diameter; ...

Sekar, Balachandran, Elumalai and Liu [5] reproduce item (1) as a numbered problem. On journal page 762, immediately after the sentence “Very recently, Khanra et al. [17] posed the following open problem:”, they print verbatim:

Problem 1. Determine the extremal values and extremal graphs with respect to Euler Sombor index over the class of all connected graphs with given diameter.

The label “Problem 1” is theirs; the problem is Khanra and Das’s. A 2026 survey by Ali, Gutman, Réti, Albalahi and Hamza [1] still poses it, as Problem 3.1.6 on journal page 109, in exactly the parameterisation used below: “Characterize graphs minimizing/maximizing the EUSO index among all fixed-order connected graphs with a given (i) diameter or (ii) domination number.”

Throughout, *given diameter* means diameter *exactly* D : for $n \geq 2$ and $D \geq 1$ put

$$\mathcal{G}_{n,D} = \{G : G \text{ connected, } |V(G)| = n, \text{ diam}(G) = D\},$$

$$M(n, D) = \max_{G \in \mathcal{G}_{n,D}} EU(G), \quad m(n, D) = \min_{G \in \mathcal{G}_{n,D}} EU(G).$$

2. RESULTS

For $a, b \geq 1$ with $a + b = n - 2$ let $H(a, b)$ be the graph on $\{u, v\} \cup W$, $|W| = n - 2$, obtained from the complete graph on W by splitting $W = A \sqcup B$ with $|A| = a$, $|B| = b$, joining u to every vertex of A and v to every vertex of B , and leaving u and v *non-adjacent*. Note $H(a, b) \cong H(b, a)$.

Theorem 1. *For every $n \geq 4$,*

$$M(n, 3) = \frac{\sqrt{3}(n-2)^2(n-3)}{2} + \sqrt{n^2 - 3n + 3} + (n-3)\sqrt{3n^2 - 15n + 19},$$

and the maximum is attained by exactly one graph up to isomorphism, namely $H(1, n-3)$.

Theorem 2. $m(9, 3) = 8\sqrt{3} + 6\sqrt{19} = 40.009800121795059\dots$, *attained by exactly one graph up to isomorphism, the bicyclic theta graph $\Theta(3, 3, 4)$ of §3. In particular the minimiser of $\mathcal{G}_{9,3}$ is neither a tree nor unicyclic: the best tree in the cell is the double star $D(3, 4)$ with $EU = \sqrt{61} + 3\sqrt{21} + 4\sqrt{31} = 43.8290342121\dots$, only third, behind the unicyclic runner-up $4\sqrt{21} + 8\sqrt{3} + 4\sqrt{7} = 42.7697144846\dots$*

At $n = 9$ Theorem 1 reads

$$M(9, 3) = 147\sqrt{3} + \sqrt{57} + 6\sqrt{127} = 329.777869165403581\dots,$$

attained uniquely up to isomorphism by $H(1, 6)$, with margin $6.5941623645\dots$ over the runner-up $EU(H(2, 5)) = 147\sqrt{3} + 2\sqrt{67} + 5\sqrt{109}$.

Theorem 1 is proved in §4 by a closed-form argument that uses nothing about n beyond $n \geq 4$ and performs no search. The uniqueness and exhaustiveness assertions of Theorem 2 are computer-assisted: the *value* $8\sqrt{3} + 6\sqrt{19}$ and the object $\Theta(3, 3, 4)$ are hand-checkable from §3, and §5 confines the minimiser to $m \leq 10$ edges by a strict inequality; but “no other member of the cell does better, and exactly one attains the value” rests on the exhaustive census of §5, shipped with this paper as `census.c`, `census.py` and `census.output.txt`.

Status. Problem 1 quantifies over all n and all D and is *not* resolved here. What is settled is the maximum half of the column $D = 3$ for all $n \geq 4$, together with the single cell $(n, D) = (9, 3)$ in both directions. The minimum half at $D = 3$ is undecided already for $n = 10$; no cell with $D \geq 4$ and $n > 9$ is touched. Every uniqueness statement is up to isomorphism. See §6.

Prior art, and what it does not settle. We make no priority claim. The extremal *object* $H(1, n - 3)$ is already in print for this class: Vetrík [8] proves sharp upper bounds for $I_f(G) = \sum_{uv \in E} f(d(u), d(v))$ over graphs of given order and diameter, and his corollary’s maximiser for the sum-connectivity index χ_a at $a = 1$ must be $H(1, n - 3)$ as well, since χ_1 is the first Zagreb index $M_1 = \sum_v d_v^2$, at $n = 9$ $M_1(H(a, b)) = 343 + a^2 + b^2$ gives $380 > 372 > 368$ on the three splits, and the whole tier $m \leq 27$ obeys $M_1 \leq \Delta \cdot 2m \leq 14m \leq 378 < 380$. What [8] does not settle, so far as we can tell, is *EU* itself: every index family named in its abstract carries an exponent ≥ 1 , whereas the edge function of (1) is 1-homogeneous and concave, and the Euler–Sombor index did not exist in 2023 [6]. That reading is an inference from an abstract and from the restatement of the hypothesis in [9], not a reading of the theorem; §6 records the risk that it is wrong. The adjacent Euler–Sombor items are these. Sekar, Balachandran, Elumalai and Liu [5] bound *EU* over the *unicyclic* graphs of given diameter, a proper subclass of $\mathcal{G}_{n,D}$, and so do not decide either extremum over the whole cell; their Theorem 3.2 is used below only as a normalisation pin. Kizilirmak [3] works with *EU* under a diameter constraint; only Lemma 2 of that paper, the value $EU(C_9)$, is used here, again as a pin. Khanra and Das [2] and the survey [1] pose the problem and do not answer it. So far as we could determine, none of these items exhibits $M(n, 3)$ or the minimiser of $\mathcal{G}_{9,3}$; we did not read [8]. What §4 supplies is the Euler–Sombor *value* of $M(n, 3)$ with an argument that the maximiser is unique, and §5 the minimum in the single cell $(9, 3)$. The auxiliary bound of Step 1 below is classical and is credited there.

3. THE TWO EXTREMAL OBJECTS, EXHIBITED

Both are printed completely, so a reader needs no code and no graph6 decoder.

The maximiser $H(1, 6)$: 9 vertices, 28 edges. Vertices $u, v, w_1, \dots, w_7$. Edges: the 21 pairs $w_i w_j$, $1 \leq i < j \leq 7$; the edge uw_1 ; and the six edges $vw_2, vw_3, \dots, vw_7$. The vertices u and v are not adjacent.

Degrees: $d(u) = 1$, $d(v) = 6$, and $d(w_i) = 7$ for $i = 1, \dots, 7$; the degree sequence is $(1, 6, 7, 7, 7, 7, 7, 7, 7)$. Checks: $21 + 1 + 6 = 28 = \binom{8}{2}$ and $1 + 6 + 7 \cdot 7 = 56 = 2 \cdot 28$.

Diameter exactly 3: the only neighbour of u is w_1 , and $w_1 \not\sim v$, so $d(u, v) = 3$ along $u w_1 w_j v$ for any $j \geq 2$; every other pair of vertices is at distance at most 2, since $w_1, \dots, w_7$ is a clique meeting every neighbourhood.

Its index: each of the 21 clique edges joins two vertices of degree 7 and contributes $\sqrt{49 + 49 + 49} = 7\sqrt{3}$; the edge uw_1 joins degrees $(1, 7)$ and contributes $\sqrt{1 + 49 + 7} = \sqrt{57}$; each edge vw_i joins degrees $(6, 7)$ and contributes $\sqrt{36 + 49 + 42} = \sqrt{127}$. Hence

$$\begin{aligned} EU(H(1, 6)) &= 147\sqrt{3} + \sqrt{57} + 6\sqrt{127} \\ &= 254.6114687126 + 7.5498344353 + 67.6165660175 \\ &= 329.7778691654 \dots \end{aligned}$$

The edge degree-pair profile is $(1, 7)^1, (6, 7)^6, (7, 7)^{21}$.

The minimiser $\Theta(3, 3, 4)$: 9 vertices, 10 edges, bicyclic. Two vertices u, v joined by three internally disjoint paths of lengths 3, 3, 4:

$$u a_1 a_2 v, \quad u b_1 b_2 v, \quad u c_1 c_2 c_3 v.$$

Vertices: $u, v, a_1, a_2, b_1, b_2, c_1, c_2, c_3$. Edges (10): $ua_1, a_1a_2, a_2v, ub_1, b_1b_2, b_2v, uc_1, c_1c_2, c_2c_3, c_3v$. Degrees: $d(u) = d(v) = 3$ and all seven internal vertices have degree 2; the degree sum is $7 \cdot 2 + 2 \cdot 3 = 20 = 2 \cdot 10$, and the cyclomatic number is $10 - 9 + 1 = 2$, so the graph is bicyclic.

Diameter exactly 3: $u \not\sim v$ and the shortest u - v path has length 3; the extremal pairs are two internal vertices on different paths, e.g. $d(a_1, b_2) = 3$ (via $a_1ub_1b_2$) and $d(a_1, c_2) = 3$ (via $a_1uc_1c_2$), and no pair exceeds 3.

Its index: the four path-interior edges $a_1a_2, b_1b_2, c_1c_2, c_2c_3$ join two degree-2 vertices and contribute $\sqrt{4 + 4 + 4} = 2\sqrt{3}$ each; the six remaining edges join a degree-2 vertex to u or v and contribute $\sqrt{4 + 9 + 6} = \sqrt{19}$ each. Hence

$$\begin{aligned} EU(\Theta(3, 3, 4)) &= 8\sqrt{3} + 6\sqrt{19} \\ &= 13.8564064606 + 26.1533936612 = 40.0098001218 \dots \end{aligned}$$

The profile is $(2, 2)^4, (2, 3)^6$.

Remark 3. All four theta graphs $\Theta(l_1, l_2, l_3)$ on 9 vertices with $l_1 + l_2 + l_3 = 10$ and every $l_i \geq 2$ share this profile and hence the *same* index $8\sqrt{3} + 6\sqrt{19}$. What separates them is the diameter: $\Theta(2, 2, 6)$, $\Theta(2, 3, 5)$ and $\Theta(2, 4, 4)$ all have diameter 4, and only $\Theta(3, 3, 4)$ has diameter 3. So the minimiser is unique *in the cell* although its value is shared by graphs outside it; the clause “diameter exactly 3” does real work.

4. PROOF OF THEOREM 1

Fix $n \geq 4$ and put $c = n - 2$. In $H(a, b)$ we have $d(u) = a$, $d(v) = b$, every vertex of W has degree $c = n - 2$, the size is $\binom{n-2}{2} + a + b = \binom{n-1}{2}$, and $\text{diam } H(a, b) = 3$ exactly (the argument of §3 for $(a, b) = (1, 6)$ is verbatim the general one).

Step 1 (edge ceiling; classical). Let $G \in \mathcal{G}_{n,3}$ and pick u, v with $d(u, v) = 3$. Put $A = N(u)$, $B = N(v)$ and $R = V(G) \setminus (\{u, v\} \cup A \cup B)$. Since $d(u, v) = 3$ we have $u \not\sim v$ and $A \cap B = \emptyset$, so $|A| + |B| + |R| = n - 2$. The pairs uv ; $\{u, b\}$ for $b \in B$; $\{u, r\}$ and $\{v, r\}$ for $r \in R$; and $\{v, a\}$ for $a \in A$ are pairwise distinct non-edges, and there are $1 + |A| + |B| + 2|R| = (n - 1) + |R| \geq n - 1$ of them. Hence

$$(2) \quad |E(G)| \leq \binom{n}{2} - (n - 1) = \binom{n-1}{2}.$$

Step 2 (equality cases). Equality in (2) forces $R = \emptyset$ and forces every pair not on the list above to be an edge, i.e. $A \cup B$ induces a complete graph, u is joined to all of A and v to all of B . Thus $G = H(|A|, |B|)$ with $|A| + |B| = n - 2$ and $|A|, |B| \geq 1$ (if $|A| = 0$ then u is isolated). Up to isomorphism these are $\lfloor (n - 2)/2 \rfloor$ graphs; at $n = 9$ exactly three, $H(1, 6)$, $H(2, 5)$, $H(3, 4)$.

Remark 4 (Attribution). Steps 1 and 2 — the bound (2) for a graph of diameter 3 and its equality characterisation — are classical: they are the Ore-type extremal problem “maximum size of a graph of given order and diameter”. They are re-derived here because the derivation is four lines, not because they are new; see [4] for a modern proof and entry point, and [7] for the 2-connected refinement. What is proved below is the Euler–Sombor layer on top: Step 3 and Step 4.

Step 3 (the whole tier $|E| \leq \binom{n-1}{2} - 1$ is beaten). If $\text{diam}(G) \geq 3$ then no vertex has degree $n - 1$ — such a vertex is adjacent to everything and forces $\text{diam} \leq 2$ — so $\Delta(G) \leq n - 2$ and every edge term of (1) obeys $\sqrt{d_x^2 + d_x d_y + d_y^2} \leq \sqrt{3\Delta^2} = \sqrt{3}(n - 2)$. (The hypothesis is $\text{diam} \geq 3$, not $\text{diam} \geq 2$: the star $K_{1, n-1}$ has diameter 2 and $\Delta = n - 1$.) Since $\binom{n-1}{2} - 1 - \binom{n-2}{2} = n - 3$, any $G \in \mathcal{G}_{n,3}$ with $|E(G)| \leq \binom{n-1}{2} - 1$ satisfies

$$EU(G) \leq \underbrace{\sqrt{3}(n - 2) \binom{n-2}{2}}_{\text{the clique part of every } H(a,b)} + \sqrt{3}(n - 2)(n - 3),$$

so that whole tier is strictly beaten by $H(1, n - 3)$ as soon as

$$(3) \quad \begin{aligned} \sqrt{3}(n - 2)(n - 3) &< \sqrt{(n - 2)^2 + (n - 2) + 1} \\ &\quad + (n - 3)\sqrt{3(n - 3)^2 + 3(n - 3) + 1}. \end{aligned}$$

Write $s = n - 3 \geq 1$; (3) becomes $\sqrt{s^2 + 3s + 3} + s\sqrt{3s^2 + 3s + 1} > \sqrt{3}s(s + 1)$. Since $\sqrt{3}s(s + 1) = s\sqrt{3s^2 + 6s + 3}$,

$$\begin{aligned} \sqrt{3}s(s + 1) - s\sqrt{3s^2 + 3s + 1} &= \frac{s(3s + 2)}{\sqrt{3s^2 + 6s + 3} + \sqrt{3s^2 + 3s + 1}} \\ &< \frac{s(3s + 2)}{2\sqrt{3s^2}} = \frac{\sqrt{3}}{2}s + \frac{1}{\sqrt{3}}, \end{aligned}$$

while $\sqrt{s^2 + 3s + 3} > s + 1$ because $s^2 + 3s + 3 - (s + 1)^2 = s + 2 > 0$. Finally

$$\begin{aligned} (s + 1) - \left(\frac{\sqrt{3}}{2}s + \frac{1}{\sqrt{3}}\right) &= \left(1 - \frac{\sqrt{3}}{2}\right)s + \left(1 - \frac{1}{\sqrt{3}}\right) \\ &= 0.1339745\dots s + 0.4226497\dots > 0, \end{aligned}$$

so (3) holds for every $s \geq 1$, with a margin growing linearly in n . The left-hand bound is valid on the *whole* discarded tier, so nothing can hide there. At $n = 9$ the cap on the tier is $27 \cdot 7\sqrt{3} = 189\sqrt{3} = 327.3576026305\dots$, below $329.7778691654\dots$.

Step 4 (which $H(a, b)$ wins). Every edge from the apex u of degree t to the clique joins degrees t and c , so those t edges contribute

$$\varphi(t) = t\sqrt{t^2 + ct + c^2}, \quad \text{and} \quad EU(H(a, b)) = \sqrt{3}c \binom{c}{2} + \varphi(a) + \varphi(c - a).$$

With $q(t) = t^2 + ct + c^2$,

$$\varphi'(t) = \frac{4t^2 + 3ct + 2c^2}{2\sqrt{q}}, \quad \varphi''(t) = \frac{8t^3 + 12ct^2 + 15c^2t + 4c^3}{4q^{3/2}} > 0$$

for $t \geq 0$ and $c > 0$, the numerator of φ'' being $2(8t + 3c)q - (4t^2 + 3ct + 2c^2)(2t + c) = (16t^3 + 22ct^2 + 22c^2t + 6c^3) - (8t^3 + 10ct^2 + 7c^2t + 2c^3)$. So φ is strictly convex, hence $a \mapsto \varphi(a) + \varphi(c - a)$ is strictly convex on $[1, c - 1]$ and is maximised only at the endpoints $a = 1$ and $a = c - 1$, which give isomorphic graphs. Therefore $H(1, n - 3)$ is the unique maximiser among the graphs of Step 2, and by Step 3 among all of $\mathcal{G}_{n,3}$. Evaluating $EU(H(1, n - 3))$ gives

$$\begin{aligned} \sqrt{3}(n - 2) \binom{n - 2}{2} + \sqrt{1 + (n - 2) + (n - 2)^2} \\ + (n - 3) \sqrt{(n - 3)^2 + (n - 3)(n - 2) + (n - 2)^2}, \end{aligned}$$

which is the closed form of Theorem 1. $\square$

5. THE MINIMUM AT $n = 9$, $D = 3$

No monotonicity is available. Adding an edge strictly increases EU , but $\mathcal{G}_{n,D}$ is closed under neither edge addition (an added edge can drop the diameter) nor edge deletion (a deleted edge can raise it or disconnect), so the minimiser is not forced onto a tree: C_6 has diameter 3 and every spanning tree of C_6 is P_6 , of diameter 5.

A floor that amputates the search space. For $x, y > 0$ we have $x^2 + xy + y^2 - \frac{3}{4}(x+y)^2 = \frac{1}{4}(x-y)^2 \geq 0$, so each edge term of (1) is at least $\frac{\sqrt{3}}{2}(d_x + d_y)$; summing over edges and applying Cauchy–Schwarz to $\sum_v d_v^2 \geq 4m^2/n$ gives, for every graph of order n and size m ,

$$(4) \quad EU(G) \geq \frac{\sqrt{3}}{2} \sum_v d_v^2 \geq \frac{2\sqrt{3}m^2}{n}.$$

At $n = 9$, $m = 11$ already gives

$$\frac{2\sqrt{3} \cdot 121}{9} = 46.5729217146 \dots > 43.8290342121 \dots = EU(D(3, 4)),$$

and the right-hand side of (4) increases in m . Since $D(3, 4)$ lies in $\mathcal{G}_{9,3}$, the minimiser has $m \in \{8, 9, 10\}$: it is a tree, unicyclic or bicyclic.

$m = 8$ by hand. The only 9-vertex trees of diameter 3 are the double stars $D(a, 7-a)$ (adjacent centres of degrees $a+1$ and $8-a$, carrying a and $7-a$ leaves), and

$$\begin{aligned} D(3, 4) &= \sqrt{61} + 3\sqrt{21} + 4\sqrt{31} = 43.8290342121 \dots, \\ D(2, 5) &= \sqrt{63} + 2\sqrt{13} + 5\sqrt{43} = 47.9355491 \dots, \\ D(1, 6) &= \sqrt{67} + \sqrt{7} + 6\sqrt{57} = 56.1301107 \dots, \end{aligned}$$

so the best tree in the cell is $D(3, 4)$, already above $8\sqrt{3} + 6\sqrt{19}$.

$m = 10$: the cheapest profile is not realisable at diameter 3. By (4) the cheapest way to spend 10 edges on 9 vertices is to keep the degrees as equal as the class allows, and the cheapest degree sequence is $(3, 3, 2^7)$. Among edge profiles realising it, the cheapest is the one with a single $(3, 3)$ -edge, worth $13\sqrt{3} + 4\sqrt{19} = 39.9522 \dots < 40.0098 \dots$; so if that profile were realisable at diameter 3 the claimed minimum would be wrong. It is not realisable. A single $(3, 3)$ -edge makes the two degree-3 vertices adjacent, which forces either a theta graph $\Theta(1, b, c)$ with $b+c=9$ or a dumbbell with a handle of length 1; and $\Theta(1, 2, 7)$, $\Theta(1, 3, 6)$, $\Theta(1, 4, 5)$ and every C_a-C_b dumbbell with $a+b=9$ have diameter at least 4. If, on the other hand, the profile has no $(3, 3)$ -edge, then the two degree-3 vertices are non-adjacent, all six of their edges run to degree-2 vertices, and a connected graph with degrees in $\{2, 3\}$ and exactly two vertices of degree 3 is a theta graph or a dumbbell; a 9-vertex dumbbell has diameter at least 5, so the graph is $\Theta(l_1, l_2, l_3)$ with $\sum l_i = 10$ and every $l_i \geq 2$, and by Remark 3 only $(3, 3, 4)$ has diameter 3. No written argument is given here for the remaining degree sequences with $m = 10$ — the next one met is $(4, 2^8)$, whose cheapest profile costs $4\sqrt{28} + 12\sqrt{3} = 41.9506 \dots$, and we do not go through the rest — and none is given for $m = 9$. Those cases are settled by the census below, not by hand.

$m = 9$, exhaustiveness and uniqueness: the census. The unicyclic case and the exhaustiveness and uniqueness assertions of Theorem 2 rest on enumeration. That enumeration is shipped with this paper: `census.c` is the

enumeration kernel, `census.py` the driver carrying the arithmetic, the controls, the automorphism counts and the verdict, and `census.output.txt` the retained output of the run, whose header records the reproduction command. It has two passes that share no enumeration machinery. Pass 1 reads every isomorphism class of connected nine-vertex graph from `nauty`'s `geng`, stratifies them by diameter, and compares EU over the cell in 80-significant-digit `Decimal` arithmetic; since `geng` emits one graph6 line per class, the number of classes in an extremal bucket *is* the number of ties, and it is 1 at each end. Pass 2 uses no generator, no canonical form and no isomorphism test: it sweeps all 2^{36} labelled graphs on nine vertices, finds no graph of the cell beating either extremum, and finds exactly $504 = 9!/|\text{Aut}(H(1,6))| = 9!/720$ labelled graphs within 10^{-6} of the maximum and exactly $90\,720 = 9!/|\text{Aut}(\Theta(3,3,4))| = 9!/4$ within 10^{-6} of the minimum, i.e. one orbit at each end and nothing else. All 43 checks pass.

The driver reproduces its controls before printing any extremum and refuses to print one if a control fails. `geng -c 9` emits $261\,080 = A001349(9)$ connected classes; the eight diameter strata are 1, 91 518, 148 229, 19 320, 1818, 180, 13, 1, which sum to that figure, so the cell $\mathcal{G}_{9,3}$ holds exactly 148 229 isomorphism classes. The 21 per-size class counts of the $D = 3$ stratum, for $m = 8, \dots, 28$,

$$\begin{aligned} &3, 17, 76, 369, 1483, 4381, 9621, 16547, 23013, 26173, 24441, \\ &18828, 12040, 6477, 2980, 1185, 412, 131, 38, 11, 3, \end{aligned}$$

sum to 148 229 as well; and the labelled pass's own count of connected labelled graphs on nine nodes, 66 296 291 072, matches the exponential formula evaluated in exact integer arithmetic. Two published values by other authors pin the normalisation (1): the census's maximum over the *unicyclic* members of $\mathcal{G}_{9,3}$ is the right-hand side of Theorem 3.2 of [5] at $n = 9$, namely $5\sqrt{57} + \sqrt{79} + \sqrt{67} + \sqrt{13} + \sqrt{19} = 62.7871695845\dots$, so that theorem is sharp at $n = 9$; and the census's minimum over $\mathcal{G}_{9,4}$ is Lemma 2 of [3], $EU(C_9) = 18\sqrt{3} = 31.1769145362\dots$, attained by the 9-cycle alone. The census also reports that the smallest gap between two distinct profile values anywhere in the cell is $1.0380112\dots \times 10^{-7}$; at 80 significant digits no comparison it makes is delicate.

Remark 5. We claim the minimiser of no cell other than $(9, 3)$.

6. WHAT IS NOT SETTLED

The one risk to Theorem 1. Vetrík [8] proves sharp *upper* bounds on I_f over graphs of given order and diameter, and §2 records that his extremal object must coincide with ours. Its full text is paywalled, with no open-access or repository copy and no preprint, and we could not read it. What we did read is the publisher's abstract, in which every index family carries an exponent ≥ 1 (χ_a , R_a , FGO_a with $a \geq 1$; $GZ_{a,b}$, $M_{a,b}$ with $a, b \geq 1$), whereas the edge function of (1) is 1-homogeneous and concave. Vetrík's

own later paper [10] says of [8] that “the result presented in [17] holds for R_a , where $a \geq 1$, and for $M_{a,b}$, where $a, b \geq 1$ ”, and the f -condition as printed in [9] fails for $f(x, y) = \sqrt{x^2 + xy + y^2}$: with $(x_1, y_1) = (1, 1)$, $(x_2, y_2) = (1, 5)$ one has $\sqrt{7} - \sqrt{3} = 0.9137\dots > \sqrt{39} - \sqrt{31} = 0.6772\dots$, so the required increasing-differences clause is violated. On that evidence *EU* appears to lie outside the class. *But that is an inference from an abstract and from a second paper’s restatement of the hypothesis, not a reading of the theorem.* If the hypothesis of [8] does admit a concave 1-homogeneous f , then Theorem 1 is a rediscovery and what survives is Theorem 2’s minimum half together with the two numerical values.

Cells left open. The minimum half at $D = 3$ is undecided for every $n \geq 10$, and every cell with $D \geq 4$ and $n > 9$ is untouched in both directions.

What the two shipped programs do and do not cover. The census of §5 — `census.c`, `census.py` and its retained transcript `census.output.txt`, whose header records the sha256 sums of the two files that were run, the wall clock and the reproduction command — is what carries the exhaustiveness and the uniqueness of both extrema of $\mathcal{G}_{9,3}$. The second program, `verify.py` with its transcript `verify.output.txt`, enumerates nothing: it re-derives what a pencil can reach — Theorem 1 in full, the two objects of §3 with their closed-form values and diameters, the strict ordering $40.0098\dots < 42.7697\dots < 43.8290\dots$ among the three named graphs $\Theta(3, 3, 4)$, one hand-built unicyclic member of the cell and the double star $D(3, 4)$, and no others, the amputation (4), the tier cap of Step 3 and the strata identities — and its closing note says correctly that it cannot itself see the exhaustiveness or the uniqueness of the minimum, and that the strata counts it re-prints are transcriptions of the census rather than re-reads. Neither program reads [8]; the f -class exclusion discussed above is a computation on the condition as printed in [9], not a reading of Vetrík’s hypothesis. Where a surd identity is asserted it is checked to a stated tolerance.

REFERENCES

- [1] A. Ali, I. Gutman, T. Réti, A. M. Albalahi, and A. E. Hamza, *Survey on extremal results and bounds for elliptic Sombor and Euler–Sombor indices*, Discrete Math. Lett. **17** (2026), 101–112, doi:10.47443/dml.2026.056. Problem 3.1.6 is on journal page 109.
- [2] B. Khanra and S. Das, *Euler Sombor index of trees, unicyclic and chemical graphs*, MATCH Commun. Math. Comput. Chem. **94** (2025), 525–548. The open problem quoted in §1 is item (1) of its concluding list, journal page 546.
- [3] G. O. Kizilirmak, MATCH Commun. Math. Comput. Chem. **94** (2025), 725–738. Its abstract, journal page 725, prints the same normalisation of *EU* as (1); the Lemma 2 used as a pin in §5 is on journal page 728. Title not recorded in the source we consulted.
- [4] *A simple proof of Ore’s theorem on the maximum size of k -connected graphs with given order and diameter*, Graphs Combin. (2023), doi:10.1007/s00373-023-02628-w. Authors not recorded in the source we consulted.
- [5] S. Sekar, S. Balachandran, S. Elumalai, and H. Liu, *Maximum Euler Sombor index of unicyclic graphs with given diameter*, MATCH Commun. Math. Comput. Chem. **95** (2026), no. 3, 759–776, doi:10.46793/match.95-3.12225. Problem 1 is on journal

- page 762; the index (1) is defined on journal page 761; Theorem 3.2 is on journal page 764.
- [6] Z. Tang, Q. Li, and H. Deng, *Int. J. Quantum Chem.* **124** (2024), e27387; and I. Gutman, *Milit. Techn. Courier* **72** (2024), 1–12. These are the two 2024 papers that introduce the Euler–Sombor index, cited as references [20] and [7] of [5]. Titles not recorded in the source we consulted.
 - [7] *Extremal 2-connected graphs with given diameter*, Tokyo J. Math. (2010), doi:10.3836/tjm/1270041609. Authors not recorded in the source we consulted.
 - [8] T. Vetrík, *Degree-based function index for graphs with given diameter*, *Discrete Appl. Math.* **333** (2023), 59–70, doi:10.1016/j.dam.2023.02.018. Nearest prior art; see §2 and §6. Its full text was not accessible to us.
 - [9] T. Vetrík and P. Kaemawichanurat, *Discrete Math. Lett.* **15** (2025), 23–30. The f -condition tested in §6 is Definition 1.1 as printed here, and its Lemma 1.1 attributes three cases to [8]. Title not recorded in the source we consulted.
 - [10] T. Vetrík, *J. Appl. Math. Comput.* (2024), doi:10.1007/s12190-024-02307-w; open access, and the source of the sentence about [8] quoted in §6. Title not recorded in the source we consulted.